

TEMPORAL VARIATIONS IN SUBLIMATION AND THEIR RELATION WITH RIDGE FORMATION IN MARTIAN MID-LATITUDE ICE SCARPS. L. S. Perez-Mancipe¹, A. M. Bramson², K. L. Laferriere³, D. M. H. Baker⁴, ¹Universidad de los Andes, Bogotá, Colombia (ls.perez2@uniandes.edu.co), ²Purdue University, West Lafayette, IN, ³University of Central Florida - Florida Space Institute, Orlando, FL, ⁴NASA Goddard Space Flight Center, Greenbelt, MD.

Introduction: The mid-latitude regions of Mars have abundant evidence for the presence of water in the form of shallowly buried ice deposits. These ice sheets undergo both accumulation and sublimation, two factors which vary over time in response to changes in Mars' orbital parameters (orbital forcing). Understanding these ice deposits thus provides a window into how the geology and climate of a planet responds to its orbital/axial changes [1].

Ice-exposing scarps were discovered in the northern and southern midlatitudes, revealing the internal structure of thick (~100 m), massive ice deposits which may be part of the latitude-dependent mantle [2]. These scarp sites are characterized by a steep (~45°) pole-facing scarp that is often adjacent to a pit with roughly triangular appearance [2–4], as shown in Figure 1. These scarps are hypothesized to form due to sublimation of the ice sheet initiating at a localized point, after which the scarp retreats equatorward, resulting in a face that is nearly perpendicular to the direction of retreat. Ridge structures within the pit are observed parallel to the present-day scarp, which we hypothesize represent periods of slower sublimation where the scarp position stagnated and accumulated debris at the base, while the intervening depressions or "valleys" correspond to episodes of rapid ice loss and retreat of the scarp position [3].

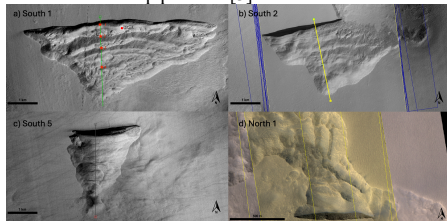


Figure 1: HiRISE images of three icy scarps in the southern hemisphere (a–c) and one in the northern hemisphere (d), with site numbers from [4]. (a) South 1: -56.6°N, 114°E. (b) South 2: -55.1°N, 109.4°E. (c) South 5: -57.2°N, 89.8°E. (d) North 1: 54.1°N 212.1°E.

We aim to test this hypothesis to understand how orbital forcing affects the geomorphology of these icy scarp sites and investigate if variable retreat rates is a plausible explanation for the observed ridges. To address this, we aim to characterize and compare ridge morphologies—such as number and spacing—across

scarps in both hemispheres and quantify the relationship between orbitally forced sublimation cycles and ridge formation.

Mapping: Using the catalog of exposed ice sites [4] as a guide, we mapped the scarp sites (8 in the South and 10 in the North) based on the number of ridges and the spacing between them using the software JMARS. We traced elevation profiles perpendicular to the scarp face, allowing us to identify the number of relative topographic highs and measure the average spacing between them. Each scarp was cataloged with its coordinates, ridge count, mean ridge separation, and brief notes on its visual appearance. In cases where Digital Terrain Models (DTMs) were not available, we relied on HiRISE imagery to visually count the number of ridges and separation. While less precise, this approach still provided a general sense of ridge scale, frequency, and clarity.

All collected data were compiled into a table to enable comparison across hemispheres. Particular attention was given to potential morphological differences between northern and southern hemisphere scarps, with an emphasis on ridge expression.

Thermal Modeling and Sublimation

Calculations: We used a 1D semi-implicit thermal conduction model that simulates the energy balance at the Martian surface and the heat transfer between numerical subsurface layers [5]. The model then calculates stability of the H₂O ice via free and forced convection for exposed ice and diffusive vapor exchange through a surficial lag deposit for subsurface ice. The model was applied to three mid-latitude scarps (South 1, 2 and 5) using present-day topography and extended over the last 20 Mya by incorporating Mars' time-varying orbital and axial parameters. Thus, the inputs varied according to the latitude, slope and orientation of each icy scarp. Cumulative ice loss is used to quantify horizontal retreat of the scarp, and temporal transitions in sublimation regimes are identified through inflection points in its derivative.

While we initially modeled the retreat of the scarps over 20 Mya, we found that the total sublimation would exceed the observed length of the scarps after just ~0.5 Mya. We thus focus our analysis on this temporal extent. With the topographic profiles of each scarp (taken from HiRISE DTMs), we qualitatively associated the ridges to the periods of slow sublimation.

Results and Discussion: We found that scarps in the southern hemisphere tend to exhibit more distinct ridges than those in the north, with average counts of 3–5 in the south compared to 2–3 in the north. Ridges in the south appear more prominent and continuous topographically, tending to run parallel to the largest pole-facing scarp face (Figures 1a–c). In contrast, northern scarps frequently display diffuse ridge patterns, and in some cases—such as North 1 (Figure 1d), North 3, and North 8—ridges appear perpendicular to the scarp. This suggests that their origin may differ or that scarp retreat may also be modulated by local heterogeneities in the ice and/or variability in local microclimate or erosional conditions.

The three southern scarp sites investigated (South 1, 2 and 5) are found to exhibit similar geomorphologies and behaviors. The lengths of the pit (assumed to be the distance of retreat, from the initiation point to present-day scarp) are 2.5–3 km. From our modeling (Figures 2–4), which accounts for evolving conditions at the scarp's location over time, we find that these features have only been retreating for the past 0.5 Mya. This age is younger than the 2.4 Mya age suggested by [3] from a constant retreat rate based on four hypothetical orbital configurations. This short timeframe also suggests that the variations in eccentricity and longitude of perihelion have a strong effect on the stability of these features, not just obliquity, and may contribute to ridge formation.

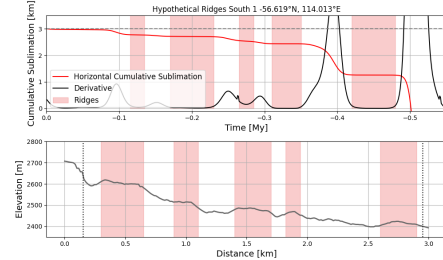


Figure 2: South 1 comparison between sublimation conditions over time (upper plot) with periods of low sublimation highlighted in red and topographic profile (lower plot) with visible ridges highlighted in red.

Figures 2, 3 and 4 show the horizontal cumulative sublimation plots over the past half million years with the periods of low sublimation highlighted in red and their topographic profiles shown in the panels below, with the locations where we interpret ridges also highlighted. We hypothesize that at these times of relative ice stability (i.e., low/no sublimation), the scarp would stabilize in place and develop ridges at the base of the cliff from accumulated mass-wasted material and overburden and/or aeolian material banked up at the

cliff base. We observed that Scarps 1 and 2 have five ridges, which aligns with the number of inflection points in sublimation conditions over the last 500 kyr, supporting the hypothesis that ridges could be related to ice stability. If our interpretation of ridges corresponding to periods of low sublimation is correct, our results therefore could allow us to place ages on the individual ridges. However, we note that some ridges in the other scarp sites are more diffuse and some are not consistent laterally along the site, which may be related to additional processes not considered by our current retreat model, such as heterogeneities within the ice deposit which could alter retreat rates besides just orbital forcing-induced climate changes. Future mapping and modeling may help elucidate these effects.

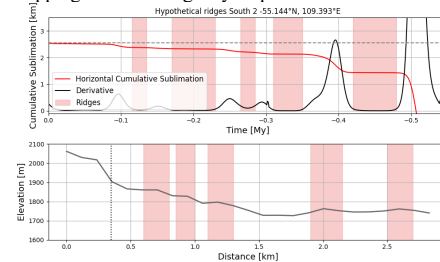


Figure 3: South 2 comparison between sublimation conditions over time (upper plot) with periods of low sublimation highlighted in red and topographic profile (lower plot) with visible ridges highlighted in red.

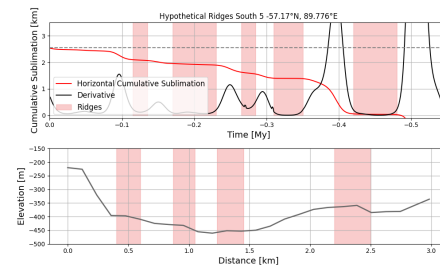


Figure 4: South 5 comparison between sublimation conditions over time (upper plot) with periods of low sublimation highlighted in red and topographic profile (lower plot) with visible ridges highlighted in red.

References: [1] Head, J. W. et al. (2003). *Nature*, 426(6968):797–802. [2] Dundas, C. M. et al. (2018). *Science*, 359(6372):199–201. [3] Williams, K. et al. (2022). *Icarus*, 386:115174. [4] Dundas, C. M. et al. (2021). *Journal of Geophysical Research: Planets*, 126(3):e2020JE006617. [5] Bramson, A. M. et al. (2017). *Journal of Geophysical Research: Planets*, 122(11), 2250–2266.

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